

Low Level Design (LLD)

Complete Course Syllabus & Topic Roadmap

71 Lessons / Topics

1. Introduction to LLD 2 Topics

- Introduction to Low Level Design
- Software Design Principles

2. SOLID Principles 5 Topics

- Single Responsibility Principle (SRP)
- Open Closed Principle (OCP)
- Liskov Substitution Principle (LSP)
- Interface Segregation Principle (ISP)
- Dependency Inversion Principle (DIP)

3. UML 2 Topics

- Unified Modeling Language (UML)
- Class UML diagrams

4. Creational Design Patterns 6 Topics

- Introduction to Design Patterns
- Singleton Design Pattern
- Factory Method
- Builder Pattern
- Abstract Factory
- Prototype Pattern

7. Multithreading and Concurrency 7 Topics

- Multithreading and Concurrency
- Creating and Managing Threads
- Thread Pools and Executors
- Thread Safety and Synchronization
- Locks and Synchronization Mechanism
- Deadlock and Prevention Techniques
- Producer Consumer Problem

8. Dependency Injection 1 Topic

- Dependency Injection

9. Exceptions and Error Handling 2 Topics

- Exception Handling (LLD)
- Building Resilient Systems

10. Best Practices in LLD 3 Topics

- All About API's
- Database Design and Integration
- How to approach a LLD Interview



5. Structural Design Patterns

7 Topics

- Adapter Pattern
- Decorator Pattern
- Facade Pattern
- Composite Pattern
- Proxy Pattern
- Bridge Pattern
- Flyweight Pattern



6. Behavioural Design Patterns 10 Topics

- Iterator Pattern
- Observer Pattern
- Strategy Pattern
- Command Pattern
- Template Method
- State Pattern
- Chain of Responsibility
- Visitor Pattern
- Mediator Pattern
- Memento Pattern



11. Interview Problems (Part-1) 10 Topics

- Parking Lot (Design)
- Parking Lot (Code)
- Logging Framework (Design)
- Logging Framework (Code)
- Traffic Signal System (Design)
- Traffic Signal System (Code)
- Vending Machine Design
- Vending Machine Code
- Task Management System Design
- Task Management System Code



12. Interview Problems (Part-2) 6 Topics

- PubSub System Design
- PubSub System Code
- ATM Machine Design
- ATM Machine Code
- Hotel Management System Design
- Hotel Management System Code



13. Interview Problems (Part-3) 10 Topics

- Elevator System Design
- Elevator System Code
- Digital Wallet Design
- Types of Locking Mechanism
- Digital Wallet Code
- Ride Booking App Design
- Ride Booking App Code
- Music Streaming Platform Design
- Streaming Protocols
- Music Streaming Platform Code